

# Maximum Mode Timing Diagrams

**Q. Discuss the Memory Read and Write timing diagrams for the 8086 microprocessor operating in Maximum Mode. How does it differ from Minimum Mode?**

## 📌 Definition to Remember

In Maximum Mode, the 8086 delegates control signal generation to the **8288 Bus Controller**. Instead of generating **RD'** or **WR'**, the CPU outputs early status codes ( **S2'**, **S1'**, **S0'** ) just before T1. The 8288 decodes these to issue the correct timing signals.

## 1. The Role of Status Signals

- **Pre-T1 Phase:** Before the T1 clock cycle even begins, the 8086 updates **S2'**, **S1'**, **S0'** to indicate the upcoming cycle (e.g., **101** for Memory Read, **110** for Memory Write).
- The 8288 uses this early warning to perfectly time the output of **ALE**, **DEN**, and commands like **MRDC'** (Memory Read).

## 2. Memory Read Timing (Maximum Mode)

- **Pre-T1:** **S2'**, **S1'**, **S0'** = **1, 0, 1** (Memory Read).
- **T1:**
  - CPU places address on AD bus.
  - **8288** pulses **ALE** high to latch the address.
  - **8288** sets **DT/R'** LOW (Receive).
- **T2:**
  - CPU removes address.
  - **8288** asserts **MRDC'** (Memory Read Command) LOW.
  - **8288** asserts **DEN** HIGH (active high) to enable transceivers.
- **T3:**
  - Memory places data on the bus. CPU reads it.
  - Status pins (**S2'**, **S1'**, **S0'**) return to **1, 1, 1** (Idle), signaling the operation is ending.
- **T4:**
  - 8288 deactivates **MRDC'** and **DEN**.

## 3. Memory Write Timing (Maximum Mode)

- **Pre-T1:** **S2'**, **S1'**, **S0'** = **1, 1, 0** (Memory Write).
- **T1:**
  - CPU places address on AD bus.
  - **8288** pulses **ALE** high.
  - **8288** sets **DT/R'** HIGH (Transmit).

- **T2:**
- CPU places data on AD bus.
- **8288** asserts **AMWC'** (Advanced Memory Write) or **MWTC'** (Memory Write) LOW.
- **8288** asserts **DEN**.
- **T3:**
- Data remains stable on bus. Status pins return to **1, 1, 1**.
- **T4:**
- 8288 deactivates **MWTC'** and **DEN**.

#### 4. Key Differences from Minimum Mode

- **Signal Source:** 8288 generates ALE, DEN, DT/R', not the CPU.
- **Command Names:** Uses **MRDC'** / **MWTC'** instead of generic **RD'** / **WR'**.
- **DEN Polarity:** DEN from 8288 is **active-high** (it was active-low **DEN'** in Min Mode).
- **Early Status:** Status pins ( **S2-S0** ) change before T1 starts.

##### ★ Must-Write Points (for 10 marks)

1. In Max Mode, the **8288 Bus Controller** handles the timing signals.
2. CPU communicates via **S2', S1', S0' status lines**, which change *before* T1 begins.
3. **T1:** 8288 generates **ALE** to latch the address put out by the CPU.
4. **T2:** 8288 asserts specific commands like **MRDC'** (Memory Read) or **MWTC'** (Memory Write) instead of generic RD/WR.
5. **DEN** is generated by the 8288 and is **active-high** in Max Mode.
6. Advanced Write Commands ( **AMWC'** ) can be used to give slow memory more time to write.
7. In T3, the status lines revert to **1, 1, 1** (Idle) to signal the end of the operation.

##### ⚡ Quick Recall

Max Mode Timing → 8288 in charge → S2,S1,S0 set before T1 → 8288 generates ALE (T1), MRDC/MWTC (T2), DEN (T2) → Status goes idle (1,1,1) in T3 → DEN is active-high